

ASSESSMENT TOOL 1

ITA0609 - Machine Learning

Sameer Ahmed G

192421154

Q1.Genetic Algorithm for Function Optimization

Problem Statement

A Genetic Algorithm (GA) is applied to maximize the function

$$f(x) = x^2$$

using an initial population consisting of the values:

2, 4, 6, 8

Determine the fitness value for each individual in the population and select the best candidates based on their fitness scores. Perform a crossover operation by exchanging the least significant bits of the selected individuals to generate new offspring. Also explain the role of mutation in improving the diversity and effectiveness of the solution.

Introduction

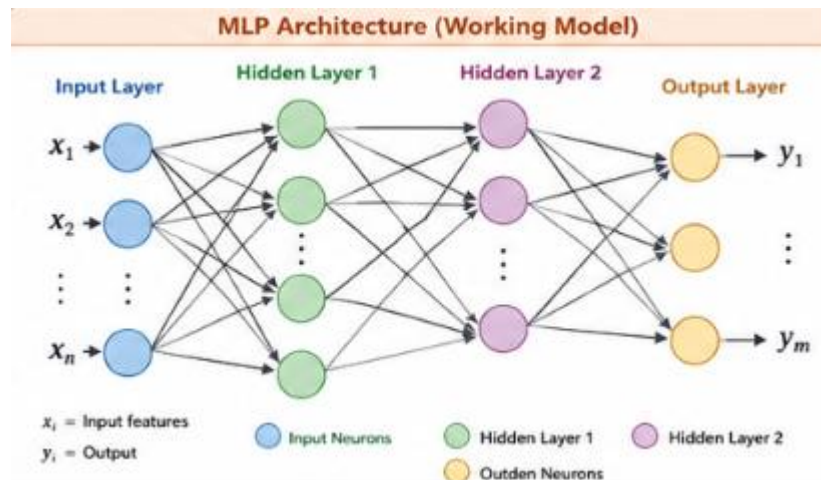
A **Genetic Algorithm (GA)** is a population-based optimization technique inspired by Charles Darwin's theory of natural evolution. It searches for the best solution by repeatedly selecting the fittest individuals and applying genetic operators such as **selection, crossover, and mutation**.

Genetic Algorithms are widely used in optimization problems where finding the optimal solution using traditional methods is difficult. Applications include scheduling, route optimization, feature selection, robotics, machine learning, and engineering design.

The objective of this problem is to maximize the function

$$f(x) = x^2$$

using the Genetic Algorithm process.



Step 1: Initial Population

The initial population consists of four individuals.

Individual Decimal Value

P_1	2
P_2	4
P_3	6
P_4	8

Step 2: Fitness Evaluation

The fitness function is

$$f(x) = x^2$$

Calculate the fitness of each individual.

Individual 1

$$f(2) = 2^2 = 4$$

Individual 2

$$f(4) = 4^2 = 16$$

Individual 3

$$f(6) = 6^2 = 36$$

Individual 4

$$f(8) = 8^2 = 64$$

The fitness table becomes

Individual x Fitness $f(x)=x^2$

P ₁	2 4
P ₂	4 16
P ₃	6 36
P ₄	8 64

Step 3: Selection

The individuals with the highest fitness are selected for reproduction.

Fitness values:

- 4
- 16
- 36
- 64

The two best individuals are

- **6 (Fitness = 36)**
- **8 (Fitness = 64)**

These become the parent chromosomes.

Step 4: Binary Representation

Convert parents into binary.

6

0110

8

1000

Step 5: Crossover

The question specifies exchanging the **least significant bit (LSB)**.

Parent 1

0110

Parent 2

1000

Exchange the last bit.

Since both parents already end with **0**, swapping the least significant bits produces no change.

Offspring 1

0110

Decimal = 6

Offspring 2

1000

Decimal = 8

Therefore,

Offspring Binary Decimal

O ₁	0110	6
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O ₂	1000	8
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Fitness:

$$6^2 = 36$$

$$8^2 = 64$$

The offspring retain the same fitness because the least significant bits were identical.

Role of Mutation

Mutation is a genetic operator that randomly changes one or more bits in a chromosome after crossover.

For example,

Original chromosome

0110

Mutated chromosome

0111

Decimal value changes from

6

to

7

New fitness

$$7^2 = 49$$

Mutation introduces new genetic material into the population, allowing the algorithm to explore better solutions that may not be obtained through crossover alone.

Advantages of Mutation

Mutation provides several benefits:

- Prevents premature convergence.
- Maintains population diversity.
- Helps escape local optimum solutions.
- Increases the probability of finding the global optimum.
- Improves exploration of the search space.

Without mutation, all chromosomes may become similar after several generations, causing the algorithm to stop improving.

Applications of Genetic Algorithms

Genetic Algorithms are widely used in:

- Machine Learning
- Neural Network Optimization
- Feature Selection
- Scheduling Problems
- Robotics
- Path Planning
- Image Processing
- Engineering Design
- Data Mining

Conclusion

The initial population contains four individuals whose fitness values are **4, 16, 36, and 64**. Based on fitness, individuals **6 and 8** are selected as parents. Since both parents have the same least significant bit, crossover does not change the offspring. Mutation plays a crucial role by introducing new variations, maintaining diversity, and improving the algorithm's ability to discover optimal solutions.

Question 2

Perceptron Model for Customer Purchase Prediction

Problem Statement

A perceptron model is used to classify whether a customer will purchase a product based on two features:

- Advertisement Exposure
- Income Level

Weights:

$$\begin{aligned}w_1 &= 0.5 \\w_2 &= 0.3\end{aligned}$$

Bias:

$$b = -0.4$$

Customer data:

Advertisement Exposure = 2

Income Level = 1

Compute the net input, apply the step activation function, and determine whether the customer purchases the product.

Introduction

A **Perceptron** is one of the simplest artificial neural network models introduced by Frank Rosenblatt. It is a binary classifier that decides whether an input belongs to one of two classes.

The perceptron computes a weighted sum of the input features, adds a bias value, and passes the result through an activation function.

It is widely used in pattern recognition, image classification, sentiment analysis, recommendation systems, and simple decision-making problems.

Given Data

Input features

Advertisement Exposure

$$x_1 = 2$$

Income Level

$$x_2 = 1$$

Weights

$$w_1 = 0.5$$

$$w_2 = 0.3$$

Bias

$$b = -0.4$$

Step 1: Calculate Net Input

The perceptron calculates

$$Net = w_1x_1 + w_2x_2 + b$$

Substitute the values

$$\begin{aligned} &= (0.5)(2) + (0.3)(1) + (-0.4) \\ &= 1.0 + 0.3 - 0.4 \\ &= 0.9 \end{aligned}$$

Therefore,

$$\boxed{Net = 0.9}$$

Step 2: Apply Step Activation Function

The Step Activation Function is

$$f(x) = \begin{cases} 1, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

Since $0.9 > 0$ The output become 1

Step 3: Interpretation

Output 1 means

Customer purchases the product.

If the output had been 0 the customer would not purchase the product.

Working of a Perceptron

The perceptron follows these steps:

1. Receive input features.
2. Multiply each feature by its weight.
3. Add all weighted values.
4. Add the bias.
5. Apply the activation function.

Flow:

Input Features

|



Multiply by Weights

|



Add Bias

|



Net Input

|



Step Activation Function

|



Purchase / No Purchase

Applications of Perceptron

Perceptrons are commonly used in:

- Email Spam Detection
- Medical Diagnosis
- Loan Approval
- Customer Purchase Prediction
- Face Recognition
- Credit Risk Analysis
- Binary Classification Problems

Advantages

- Simple and easy to implement.
- Fast computation.
- Suitable for binary classification.
- Low computational cost.
- Easy to interpret.

Limitations

- Works only for linearly separable data.
- Cannot solve XOR-type problems.
- Limited learning capability.
- Less powerful than multi-layer neural networks.

Conclusion

The perceptron computes the weighted sum of the input features and bias, resulting in a **net input of 0.9**. Since the value is positive, the step activation function outputs **1**, indicating that the **customer will purchase the product**. The perceptron provides a simple yet effective method for solving binary classification problems in machine learning.